

TECHNICAL MEMORANDUM:
IES METHOD FOR DETERMINING
CORRELATED COLOR TEMPERATURE
(CCT) AND DISTANCE FROM THE
PLANCKIAN LOCUS OF LIGHT SOURCES
AN AMERICAN NATIONAL STANDARD

**TECHNICAL MEMORANDUM:
IES METHOD FOR DETERMINING CORRELATED
COLOR TEMPERATURE (CCT) AND DISTANCE
FROM THE PLANCKIAN LOCUS OF LIGHT SOURCES
AN AMERICAN NATIONAL STANDARD**

Publication of this Technical Memorandum
has been approved by IES.

Suggestions for revisions
should be directed to IES.



Copyright 2024 by the Illuminating Engineering Society.

Approved by the IES Standards Committee February 5, 2024, as a Transaction of the Illuminating Engineering Society.

Approved May 20, 2024, as an American National Standard.

All rights reserved. No part of this publication may be reproduced in any form, in any electronic retrieval system or otherwise, without prior written permission of the IES.

Published by the Illuminating Engineering Society, 85 Broad Street, New York, New York 10004.

IES Standards are developed through committee consensus and produced by the IES Office in New York. Careful attention is given to style and accuracy. If any errors are noted in this document, please forward them to the Director of Standards, at standards@ies.org or the above address, for verification and correction. The IES welcomes and urges feedback and comments.

ISBN 978-0-87995-476-5.

Printed in the United States of America.

DISCLAIMER

IES publications are developed through the consensus standards development process approved by the American National Standards Institute. This process brings together volunteers representing varied viewpoints and interests to achieve consensus on lighting recommendations. While the IES administers the process and establishes policies and procedures to promote fairness in the development of consensus, it makes no guaranty or warranty as to the accuracy or completeness of any information published herein.

The IES disclaims liability for any injury to persons or property or other damages of any nature whatsoever, whether special, indirect, consequential or compensatory, directly or indirectly resulting from the publication, use of, or reliance on this document.

In issuing and making this document available, the IES is not undertaking to render professional or other services for or on behalf of any person or entity. Nor is the IES undertaking to perform any duty owed by any person or entity to someone else. Anyone using this document should rely on his or her own independent judgment or, as appropriate, seek the advice of a competent professional in determining the exercise of reasonable care in any given circumstances.

The IES has no power, nor does it undertake, to police or enforce compliance with the contents of this document. Nor does the IES list, certify, test or inspect products, designs, or installations for compliance with this document. Any certification or statement of compliance with the requirements of this document shall not be attributable to the IES and is solely the responsibility of the certifier or maker of the statement.

AMERICAN NATIONAL STANDARD

Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria have been met by the standards developer.

Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

The use of American National Standards is completely voluntary; their existence does not in any respect preclude anyone, whether that person has approved the standards or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standards.

The American National Standards Institute does not develop standards and will in no circumstances give an interpretation to any American National Standard. Moreover, no person shall have the right or authority to issue an interpretation of an American National Standard in the name of the American National Standards Institute. Requests for interpretations should be addressed to the secretariat or sponsor whose name appears on the title page of this standard.

CAUTION NOTICE: This American National Standard may be revised at any time. The procedures of the American National Standards Institute require that action be taken to reaffirm, revise, or withdraw this standard no later than five years from the date of approval. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute.

Prepared by the IES Color Committee.

Tony Esposito, *Co-chair*

Jason Livingston, *Co-chair*

Members

Jess Baker

Ute C. Besenecker

Eric Bretschneider

Aurelien David

Kevin W. Houser

Gabi Korac

Wendy Luedtke

Rez Mani

Yoshi Ohno

Paul K. Pickard

Holly R. Ratafia

Michael P. Royer

Paul Scheidt

Kevin A.G. Smet

Minchen Wei

Lorne Whitehead

Advisory Members

Nicola Bortolaso

John D. Bullough

Golshan Coleiny

Alp Durmus

Frank A. Florentine

Kasey Holland

Kwong H. Man

Michael Murdoch

Abhijit Sarkar

Hugo Solano

Mike Wood

CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction.....	1
1.2	Scope.....	1
2.0	Calculation Procedure.....	2
2.1	LUT Generation.....	2
2.2	LUT Search and Interpolation	2
2.3	Duv Calculation	3
3.0	Performance.....	3
4.0	Use and Limitations	3
	Annex A – Lookup Table.....	4
	Annex B – Calculation of the Slope of an Isotemperature Line	5
	Annex C – Use of the Recommended Method With Alternative Chromaticity Diagrams.....	5
	Annex D – Pseudocode for Fibonacci Search	6
	References	8

1.0 Introduction and Scope

1.1 Introduction

Correlated color temperature (CCT) is an important metric that has been used for decades to engineer and specify light sources. In recent years, it has been increasingly common to augment CCT with distance from the Planckian locus (D_{uv} or Duv), which allows for a complete specification of a light source's chromaticity and a better indication of the relative color appearance of the light. Despite being clearly defined as “the absolute temperature of a blackbody whose chromaticity most nearly resembles that of the light source,”¹ or the “temperature of a Planckian radiator having the chromaticity nearest the chromaticity associated with the given spectral distribution” on the 1960 UCS, where (u , v) are the coordinates of the Planckian locus and the test stimulus,² there is no exact way to calculate CCT. It requires a highly accurate approximation, which can be accomplished in many ways.

More than a dozen methods for calculating a value for CCT were proposed and documented in scientific literature between the 1930s and 2023.^{3–20} These methods have utilized a wide variety of approaches, including lookup tables with various interpolation formulas, equation-based methods reliant on modelling elements of a chromaticity diagram, and several types of iterative methods. The error for each method can be determined for a range of test chromaticity coordinates with known CCT values, and past evaluations have shown that errors can range from less than a thousandth of a kelvin to several hundred kelvins,^{10,18,21–23} depending on the exact implementation.

Prior to the publication of this Technical Memorandum, no method for calculating CCT (nor Duv) has been formalized by a relevant lighting standards organization. This omission may have occurred because almost all the previously identified methods have suitable speed and accuracy for commercial applications (e.g., photometric testing), which usually only require a single calculation, so the speed does not matter, and small differences in CCT are perceptually irrelevant.

However, in some cases an accurate calculation method with known precision is required, such as when precise CCT values of reference illuminants are calculated for standards documents and when results need to be reproduced to several significant digits. Further, some situations where CCT is calculated require both fast and efficient calculations—for example, during optimizations where millions or billions of CCT calculations may be performed, or in control hardware providing real-time information on CCT. Finally, having a single, standardized method for calculating CCT and Duv that is widespread in its use is advantageous for providing consistency and eliminating uncertainty.

1.2 Scope

This document formalizes a recommended method, for the purposes described in **Section 1.1**, for computing the CCT and Duv of a light source (or theoretical illuminant) given the input of its chromaticity coordinates, which are typically calculated from a spectral power distribution. It provides all necessary data and formulae to perform calculations and provides a summary of recent work that justifies the choice of this method. It also provides a brief explanation of the use and limitations of CCT and Duv.

The recommended method is a modification¹² of a method first published by Robertson in 1968.¹⁷ The main advancement with respect to the original is the introduction of a finer-resolution lookup table (LUT; see **Annex A**), which substantially improves the accuracy of the method. The recommended method, hereafter referred to as the IES method, uses an increment of 1% for each row of the table, resulting in an LUT that has 372 rows covering the range of 1000 to 40,000 K, instead of the original method, which had 31 rows covering 1667 K to 100,000 K. The same interpolation method is utilized, with a small correction to prevent errors around 1635 K, where the sign of the slope of the isotherm lines changes. Recommendations are provided to maximize the efficiency of the calculation, including precomputing key values that can accompany the main LUT.

This recommended method has the following features and benefits:

1. CCT error of less than 0.1 K over the range of 1000 to 40,000 K²³ (see **Section 3**)

2. Duv error less than 0.0001 over the range of 1000 to 40,000 K²³ (see **Section 3**)
3. More computationally efficient (shorter calculation time) than other methods currently available that achieve comparable performance²³
4. Requires minimal data storage (approximately 70 kB in CSV file format)¹²
5. Straightforward calculations that are easy to replicate in different implementations
6. Easily adapted, via recalculation of the LUT, if the color matching functions or uniform chromaticity scale (UCS) diagram used to determine CCT were to be updated

2.0 Calculation Procedure

The method has three phases: LUT generation; LUT search and interpolation to determine CCT; and calculation of Duv. It is only necessary to perform the LUT generation portion once, as the LUT values do not change based on the light for which the CCT is being calculated. It is provided in **Annex A**, but the generation procedure is also described here for thoroughness.

2.1 LUT Generation

Minimally, the LUT includes sequential values for temperature (of a Planckian radiator) and corresponding CIE 1960 (u , v) chromaticity coordinates. These can be calculated according to CIE 15:2018.²⁴ The standard 372-row LUT specified for this method ranges from 1000 K to nominally 40,000 K with an increment of 1%, and should be calculated to 15 decimal places as provided in **Annex A**. That is, the first row corresponds to a temperature of 1000 K, the second a temperature of $1000 \times 1.01 = 1010$ K, the third $1010 \times 1.01 = 1020.1$ K, and so forth to a maximum of 40,107.8769032694 K. To maintain the accuracy levels of the specified method, all values in the LUT should be computed with 15 significant digits (or 64-bit double precision). If the range of interest is smaller than 1000 K to approximately 40,000 K, a truncated LUT may be used, but the temperature values are required to correspond to those calculated beginning with 1000 K and increasing with a 1% increment; otherwise, the

determined CCT values will not match exactly with those calculated using the larger (recommended) LUT.

To improve the speed of subsequent calculations, several quantities used in the method can be pre-computed for each combination of temperature, u coordinate, and v coordinate that comprise a row of the most basic LUT—index value j . These values are related to the slope (m) of the isotherm line at the given temperature, and the temperature (T):

- m_j
- $1/(1 + m_j^2)^{0.5}$
- $1/T_j$
- $1/T_{j+1} - 1/T_j$
- $m_j \times m_{j+1}$

Steps for computation of m_j are provided in **Annex B**, and all values are provided in **Annex A**.

With these five quantities, the standard LUT becomes 8 columns by 372 rows. It is approximately 70 kB in CSV format but can also be programmed directly into the calculation program.

It is important to note that while current definitions of CCT and Duv specify that they be calculated in the CIE 1960 (u , v) chromaticity diagram, it is possible to implement this method for other chromaticity diagrams by recalculating the LUT, as described in **Annex C**.

2.2 LUT Search and Interpolation

Given the LUT, the next step is to find the two rows corresponding to the closest points to the (u , v) chromaticity of the light source in question, which is referred to as the *test coordinate*. The test coordinate should be calculated from a light source spectral power distribution with as high a precision as possible—rounding values for (u , v) coordinates can increase the calculation error. Reduced precision (i.e., fewer decimal places) may reduce the calculation accuracy such that errors exceed the specified thresholds.

Searching can be done most efficiently with a Fibonacci search sequence^{12,25,26} or with a binary search, although any search technique will yield the same two rows, including a linear search from the beginning of the

LUT. Pseudocode for a Fibonacci search is provided in **Annex D**.

Once the two rows are identified, **Equation 2-1** is used to interpolate the CCT value (T_{cp}).

$$T_{cp} = \left[\frac{1}{T_j} + \frac{|d_j|}{|d_j| - |d_{j+1}|} \left(\frac{1}{T_{j+1}} - \frac{1}{T_j} \right) \right]^{-1}, \quad (2-1)$$

where:

j and $j+1$ are the index values of the two LUT rows corresponding to the closest points to the test coordinates

d is the distance between the test coordinates and the nearest isotherm line

T is the temperature of the specified row

It is important to note that **Equation 2-1** is slightly modified from the original version proposed by Robertson in 1968.¹⁷ The use of absolute values for the distance terms ameliorates potential errors due to the slope of the isotherm line changing from positive to negative. Without this modification, errors around 1625 K—outside the range originally considered in 1968—would be slightly inflated.

2.3 Duv Calculation

Once the CCT is known, Duv (symbol D_{uv}) is calculated as the Euclidian distance between the test chromaticity coordinates ($u_{test,Tcp}$, $v_{test,Tcp}$) and coordinates of a Planckian radiator of the same CCT ($u_{Planckian,Tcp}$, $v_{Planckian,Tcp}$), as shown in Equation 2-2. It is important that the latter be recalculated at the specific CCT, rather than using the value of the closest Planckian radiator in the LUT.

$$D_{uv} = \sqrt{(u_{test,Tcp} - u_{Planckian,Tcp})^2 + (v_{test,Tcp} - v_{Planckian,Tcp})^2} \quad (2-2)$$

3.0 Performance

This TM follows recent work comparing the speed and accuracy of various CCT calculation methods. In that analysis, the IES method had a maximum CCT error of 0.097 K and a maximum Duv error of 0.0000012 for a set

of 1,270 reference chromaticity coordinates. Notably, these error levels are attainable from other methods that can be adjusted by increasing the resolution of the LUT or that are iterative in nature and reach a preset stopping criterion.

A key advantage of the IES method is that it was found to be the most computationally efficient method that could achieve a maximum CCT error less than 0.1 K and a maximum Duv error of less than 0.0001, which are approximately an order of magnitude smaller than typically reported. While absolute calculation time is dependent on programming efficiency and computing power, recent research²³ showed the time for each conversion of chromaticity coordinates to CCT and Duv to be 80 nanoseconds when programmed in C and run on an institutional computer, and approximately 1.5 milliseconds when programmed in Python and run on a typical office laptop computer.

4.0 Use and Limitations

Regardless of calculation method, the use and limitations of CCT have been well documented.^{e.g.,27} Key items to remember when using CCT and Duv include:

- CCT and Duv are required to be used together for a complete specification of chromaticity.¹⁶ Either alone is insufficient for characterizing even the relative appearance of a light stimulus. That is, with all other factors being equal, two lights with the same CCT can vary in appearance; this variation is generally in the pink-green dimension. Likewise, light sources with the same Duv value can vary in the blue-yellow dimension.
- CCT and Duv (and chromaticity) do not convey absolute color appearance; the same stimulus with the same CCT and Duv can appear very differently if, for example, the surround condition is different or the state of adaptation of the observer is different.¹⁶ For example, a light with CCT of 4000 K and Duv of 0.0000 may appear cool blue next to one with CCT of 2700 K, but warm next to one with CCT of 6500 K.
- CCT and Duv reduce the complex information contained in a spectral power distribution to a pair

of numbers. They should not be used as a proxy for other quantities derived from a spectral power distribution (e.g., alpha-opic flux),²⁸ especially when accuracy of calculations is important. For example, melanopic flux for lights with the same CCT, Duv, and luminous flux can vary by a factor of 2 or more.

- There is not a universally preferred CCT and Duv or single nominally white CCT and Duv,^{29–32} and older findings regarding an illuminance-CCT preference band have been discredited.³³ Many factors, such as spectral power distribution, culture, geography, adaptation state, and personal preferences can affect preferred chromaticity.
- The CIE system of colorimetry, on which CCT and Duv calculations rely, is based on standard observers and does not account for interobserver variability. This variability can lead to different real observers having different perception of “matches” between lights characterized as having the same chromaticity (i.e., the same CCT and Duv).^{34,35}

This Technical Memorandum and the IES method for calculating CCT and Duv do not address any of these issues. Nonetheless, they represent an important advancement because they provides a formalized method for calculating CCT and Duv, which will allow for consistent application in the lighting profession. The method is suitably accurate and computationally efficient, thus facilitating optimizations and real-time calculations in addition to other one-off calculations. Therefore, it is recommended for widespread use.

If the method described in this Technical Memorandum is not used, alternative methods should minimally achieve the same maximum error thresholds: 0.1 K for CCT and 0.0001 for Duv. These can be achieved by several calculation methods, and the specific method implemented should be noted. Alternative methods are likely to be less computationally efficient and require more time per calculation to achieve the same accuracy.

Annex A Lookup Table

Note: This annex is not part of ANSI/IES TM-40-24. It is intended for information only.

Index	Color Temperature	u	v	m_j	$1/(1 + m_j^2)^{0.5}$	$1/T_j$	$1/T_{j+1} - 1/T_j$	$m_j * m_{j+1}$
1	1000.0000000000	0.448010894640648	0.354624980858124	11.53452515075300	0.08637225382345	0.00100000000000	-0.00000990099010	134.079000493862
2	1010.0000000000	0.445681292405968	0.354826175721752	11.62414566195720	0.08571124699541	0.00099009900990	-0.00000980296049	136.225509482143
3	1020.1000000000	0.443353478205085	0.355025626391888	11.71918465612260	0.08502119999877	0.00098029604941	-0.00000970590148	138.520887813795
4	1030.3010000000	0.441027834601749	0.355223233656728	11.82001068149610	0.08430113563075	0.00097059014793	-0.00000960980344	140.977553562295
5	1040.6040100000	0.438704738259149	0.355418897533421	11.92702421013800	0.08355005940775	0.00096098034448	-0.00000951465688	143.609258391070
6	1051.01005010000	0.436384559838273	0.355612517300679	12.04066126310040	0.08276695999222	0.00095146568761	-0.00000942045235	146.431268332494
7	1061.52015060100	0.434067663905223	0.355803991533481	12.16139754560190	0.08195080965621	0.00094204523525	-0.00000932718055	149.460574129875
8	1072.13535210701	0.431754408847424	0.355993218139892	12.28975317758000	0.08110056478149	0.00093271805471	-0.00000923483222	152.716136826410
9	1082.85670562808	0.429445146798627	0.356180094400057	12.42629812167480	0.08021516639607	0.00092348322248	-0.00000914339824	156.219175548753
10	1093.68527268436	0.427140223572579	0.356364517007382	12.57165843110300	0.07929354074777	0.00091433982424	-0.00000905286955	159.993506020297

The full table can be found at https://media.ies.org/docs/standards/TM-40-24_AnnexA.xlsx (Accessed 2024 May 29).

Annex B

Calculation of the Slope of an Isotemperature Line

Note: This annex is not part of ANSI/IES TM-40-24. It is intended for information only.

According to CIE 15:2018,²⁴ the spectral power distribution of a Planckian radiator at temperature (T) is calculated as shown in **Equation B-1**.

$$M(\lambda, T) = c_1 \lambda^{-5} (e^{c_2/(\lambda T)} - 1)^{-1}, \quad (\text{B-1})$$

where:

$$c_1 = 3.741832 \times 10^{-16} \quad (\text{in watts per square meter, W/m}^2)$$

$$c_2 = 1.4388 \times 10^{-2} \quad (\text{in millikelvins, mK})$$

The unscaled tristimulus values of the Planckian radiator can then be calculated as shown in **Equations B-2** through **B-4**.

$$X(T) = \int M(\lambda, T) \bar{x}(\lambda) d\lambda \quad (\text{B-2})$$

$$Y(T) = \int M(\lambda, T) \bar{y}(\lambda) d\lambda \quad (\text{B-3})$$

$$Z(T) = \int M(\lambda, T) \bar{z}(\lambda) d\lambda \quad (\text{B-4})$$

These values are integrated over the visible range. In practice, these values are calculated as the sum product; for the table in **Annex A**, the range of 360 to 830 nm was used, with an increment of 1 nm. The first derivative of a Planckian radiator at temperature (T) is calculated as shown in **Equations B-5** and **B-6**:

$$M'(\lambda, T) = \frac{\partial M(\lambda, T)}{\partial T} = c_3(T) \lambda^{-6} (e^{c_2/(\lambda T)} - 1)^{-2} e^{c_2/(\lambda T)}, \quad (\text{B-5})$$

$$\text{with } c_3 = \frac{c_1 c_2}{T^2} \quad (\text{B-6})$$

With this, the first derivatives of the tristimulus values can be calculated as shown in **Equations B-7** through **B-9**.

$$X'(T) = \int M'(\lambda, T) \bar{x}(\lambda) d\lambda \quad (\text{B-7})$$

$$Y'(T) = \int M'(\lambda, T) \bar{y}(\lambda) d\lambda \quad (\text{B-8})$$

$$Z'(T) = \int M'(\lambda, T) \bar{z}(\lambda) d\lambda \quad (\text{B-9})$$

From this, **Equations B-10** and **B-11** can be calculated.

$$u'(T) = \frac{4X'(T)[X(T) + 15Y(T) + 3Z(T)] - 4X(T)[X'(T) + 15Y'(T) + 3Z'(T)]}{[X(T) + 15Y(T) + 3Z(T)]^2} \quad (\text{B-10})$$

$$v'(T) = \frac{6Y'(T)[X(T) + 15Y(T) + 3Z(T)] - 6Y(T)[X'(T) + 15Y'(T) + 3Z'(T)]}{[X(T) + 15Y(T) + 3Z(T)]^2} \quad (\text{B-11})$$

The slope of the isotemperature line is then calculated as shown in **Equation B-12**.

$$m_i = -u'(T)/v'(T) \quad (\text{B-12})$$

Annex C

Use of the Recommended Method With Alternative Chromaticity Diagrams

Note: This annex is not part of ANSI/IES TM-40-24. It is intended for information only.

The CCT calculation method described in this Technical Memorandum can be applied to other chromaticity diagrams, including those generated from different color matching functions. The only action necessary is to recalculate the LUT using new color matching functions (or cone fundamentals) and/or projective transformation. For example, the method could be used to calculate CCT_{st} and Dst,³⁶ which are recently proposed analogs to CCT and Duv in the (s , t) chromaticity diagram built from the CIE 2015 cone-fundamental based tristimulus functions. If any alternative color matching functions or projective transformation is used, the results should be clearly noted to avoid confusion with the specifically defined CCT and Duv quantities.

Annex D

Pseudocode for Fibonacci Search

Note: This annex is not part of ANSI/IES TM-40-24. It is intended for information only.

The standard LUT has j rows indexed from 1 to 372. The temperature, u coordinate, and v coordinate can be denoted as t_j , u_j , and v_j for the j^{th} row, with j in $[1, 372]$. It should be noted that this discussion focuses on the three primary columns of the LUT.

Given chromaticity coordinates (u_x, v_x) , the goal is to find the index, j , for which the distance from (u_x, v_x) to (u_j, v_j) is a minimum. It suffices to find the minimum square of the distance between (u_x, v_x) and (u_j, v_j) , as $a^2 < b^2$ implies $|a| < |b|$. This eliminates needing to take square roots for each distance computed.

For convenience, the squared distance can be defined as the function g as shown in **Equation C-1**.

$$g(u_j, v_j, u_x, v_x) = (u_j - u_x)^2 + (v_j - v_x)^2 \quad (\text{C-1})$$

An additional requirement for the Fibonacci search technique is an array of Fibonacci numbers, $F[i]$, that omits the traditional first two elements of the sequence, 0 and 1, and starts instead with next two elements 1 and 2 as shown in **Equations C-2** through **C-4**.

$$F[0] = 1 \quad (\text{C-2})$$

$$F[1] = 2 \quad (\text{C-3})$$

$$F[i] = F[i - 1] + F[i - 2], i \geq 2 \quad (\text{C-4})$$

The length of F , $l + 1$, is chosen such that **Equations C-5** and **C-6** hold:

$$F[l - 1] < 372 \quad (\text{C-5})$$

$$F[l] \geq 372 \quad (\text{C-6})$$

Then during each stage of the search, four indices are tracked into the table: A *left* and a *right* index that define the limits of the index range in which the point of minimum distance from the target lies; and two indices a and b that lie between *left* and *right* and are successive Fibonacci numbers after *left*. Each

step of the search then involves reducing the distance between *left* and *right* to the next lower Fibonacci number by replacing one of the endpoints with the interior point at largest distance from the target and choosing a new interior point that maintains the relationships in **Equations C-7** through **C-9**, with k decreasing at each stage.

$$a = \text{left} + F[k] \quad (\text{C-7})$$

$$b = \text{left} + F[k + 1] \quad (\text{C-8})$$

$$\text{right} = \text{left} + F[k + 2] \quad (\text{C-9})$$

It is important to note that when the row j closest to the target is greater than $F[l - 1]$ as defined in **Equations C-5** and **C-6** and $F[l]$ is strictly greater than $j - 1$, **Equations C-7** through **C-9** may generate indices greater than $n - 1$; thus, it is important that additional restrictions be placed on a , b , and *right* to avoid referring to nonexistent rows of the LUT, and to maintain strict inequality of a and b as defined in **Equations C-10** through **C-12**.

$$\text{right} \leq 372 \quad (\text{C-10})$$

$$b \leq 372 \quad (\text{C-11})$$

$$a < b \quad (\text{C-12})$$

The *left* and *right* indices are initialized to 0 and 372, respectively, and then only set to successive a or b values; hence, **Equation C-10** always holds if **Equation C-11** is enforced. As b is updated, **Equation C-11** will have to be enforced, and as a is updated, **Equation C-12** will have to be enforced. It is important to note that when j is 1, then *left*, a , and b necessarily point to consecutive rows in the LUT, and *right* is either one more than b or equal to b if the solution is at the very end of the table. Thus, the minimum-distance row from the table will be either a or b if *left* is not 1 and *right* is not 372. The latter conditions are ensured by extending the LUT downward and upward so that (u_1, v_{01}) and (u_{372}, v_{372}) are never the closest points to a target (u_x, v_x) .

A pseudo-code algorithm for the Fibonacci search as described is given in **Figure C-1**. The variables *dsq_left*, *dsq_right*, *dsq_a*, and *dsq_b* are used to hold the distances from (u_x, v_x) to the points in the corresponding rows of the table so as to compute the distance to any given row in the table only once.

Technical Memorandum: IES Method for Determining Correlated Color Temperature (CCT) and Distance from the Planckian Locus of Light Sources

```

left = 1; 1
right = 372; 2
a = F[l - 2]; 3
b = F[l - 1]; 4
dsq_left = g(u1, v1, ux, vx); 5
dsq_a = g(ua, va, ux, vx); 6
dsq_b = g(ub, vb, ux, vx); 7
dsq_right = g(uj-1, vj-1, ux, vx); 8
for (i = l - 3; i ≥ 0; i = i - 1) { 9
    if (dsq_a < dsq_b) { 10
        /* row a was closer shift to [left, b] */ 11
        right = b; 12
        dsq_right = b; 13
        b = a; 14
        dsq_b = dsq_a; 15
        a = left + F[j]; 16
        if (a > b) { 17
            a = b - 1; 18
        } 19
        dsq_a = g(ua, va, ux, vx) 20
    } else { 21
        /* row b was closer, shift to [a, right] */ 22
        left = a; 23
        dsq_left = dsq_a; 24
        a = b; 25
        dsq_a = dsq_b; 26
        b = left + F[i + 1]; 27
        if (b ≥ (right)) { 28
            b = (right - 1); 29
        } 30
        dsq_b = g(ub, vb, ux, vx); 31
    } 32
} 33
if (dsq_a < dsq_b) { 34
    j = a; 35
} else { 36
    j = b; 37
} 38
/* j is the index of the row of the table closest to (ux, vx) */ 39

```

Figure C-1. Pseudocode for Fibonacci search of the standard LUT.

It should be noted that only one of line 32 or line 20 is executed; thus, only one distance computation (call to *g*) is made per iteration of the loop. In practice, the LUT may have a different number of columns depending on the method of interpolation to be used, and it is useful to store it in a single linear array of length $m * n$, where m is the number of columns. Then it saves a little bit of indexing time to have the search generate $k * m$ instead of k , so as to get the offset of the first element of the j^{th} row of the table. This is easily accomplished by starting with the

Fibonacci array generation replacing **Equations C-2** and **C-3** with **Equations C-13** and **C-14**.

$$F[1] = m; \quad (\text{C-13})$$

$$F[2] = m + m; \quad (\text{C-14})$$

Then, **Equation C-4** can be used to generate the rest of the table until $F(l) \geq m * (372)$. Also, line 18 in **Figure C-1** needs to change to $a = b - m$, line 29 needs to change to $b = \text{right} - m$, and the (372) in lines 28 and 29 of **Figure C-1** (i.e., *right*) needs to change to $m * (372)$.

REFERENCES

1. Illuminating Engineering Society. ANSI/IES LS-1-22: Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2022.
2. International Commission on Illumination (CIE). International Lighting Vocabulary. CIE S017/E:2020. Vienna: CIE; 2020.
3. Davis R. A correlated color temperature for illuminants. *Nat Bur Stand J Res.* 1931;7:659.
4. Judd DB. Estimation of chromaticity differences and nearest color temperature on the standard 1931 ICI colorimetric coordinate system. *J Res National Bur Standards.* 1936;17:771.
5. Guo X, Houser KW. A review of colour rendering indices and their application to commercial light sources. *Light Res Technol.* 2004;36:183-99.
6. Xingzhong Q. Formulas for computing correlated color temperature. *Color Res Appl.* 1987;12:285-7.
7. Hernández-Andrés J, Lee RL, Romero J. Calculating correlated color temperatures across the entire gamut of daylight and skylight chromaticities. *Appl Opt.* 1999;38:5703.
8. McCamy CS. Correlated color temperature as an explicit function of chromaticity coordinates. *Color Res Appl.* 1992;17:142-4.
9. Gardner JL. Correlated colour temperature — uncertainty and estimation. *Metrologia.* 2000;37:381-4.
10. Li C, Cui G, Melgosa M, Ruan X, Zhang Y, Long M, Xiao K, Luo MR. Accurate method for computing correlated color temperature. *Opt Express.* 2016;24:14066.
11. Zhang F. High-accuracy method for calculating correlated color temperature with a lookup table based on golden section search. *Optik (Stuttg).* 2019;193:163018.
12. Baxter D, Royer M, Smet K. Modifications of the Robertson method for calculating correlated color temperature to improve accuracy and speed. *Leukos.* Epub ahead of print 2023. DOI: 10.1080/15502724.2023.2166060.
13. Harding HGW. The colour temperature of light sources. *Proc Phys Soc Sect B.* 1950;63:685-98.
14. MacAdam DL. Correlated color temperature? *J Opt Soc Am.* 1977;67:839.
15. Kelly KL. Lines of constant correlated color temperature based on MacAdam's (u, v) uniform chromaticity transformation of the CIE diagram. *JOSA.* 1963;53:999-1002.
16. Wyszecki G, Stiles WS. *Color science*, Vol. 8. New York: Wiley, 1982.
17. Robertson AR. Computation of correlated color temperature and distribution temperature. *J Opt Soc Am.* 1968;58:1528.
18. Ohno Y. Practical use and calculation of CCT and Duv. *Leukos.* 2014;10:47-55.
19. Schanda J, Mészáros M, Czibula G. Calculating correlated color temperature with a desktop programmable calculator. *Color Res Appl.* 1978;3:65-9.
20. Krystek M. An algorithm to calculate correlated colour temperature. *Color Res Appl.* 1985;10:38-40.
21. Prytkov S V., Kolyadin M V. Error estimation for the methods of correlated colour temperature calculation. *Light Eng.* 2021;29:70-7.

Technical Memorandum: IES Method for Determining Correlated Color Temperature (CCT) and Distance from the Planckian Locus of Light Sources

22. Zheleznikova OE. Selection of optimal method of correlated colour temperature calculation. *J Mech Contin Math Sci.* 2020;spl8:134-43.
23. Smet KAG, Royer MP, Baxter D, Bretschneider E, Esposito T, Houser K, et al. Recommended method for determining the correlated color temperature and distance from the Planckian locus of a light source. *Leukos.* Submitted.
24. International Commission on Illumination (CIE). *CIE 15:2018, Colorimetry*, 4th ed. Vienna: CIE; 2018.
25. Overholt KJ. Efficiency of the Fibonacci search method. *BIT Numer Math.* 1973;13:92-6.
26. Avriel M, Wilde DJ. Optimally proof for the symmetric fibonacci search technique. *Fibonacci Q J.* 1966;265-9.
27. Durmus D. Correlated color temperature: Use and limitations. *Light Res Technol.* 2022;54:363-75.
28. Esposito T, Houser K. Correlated color temperature is not a suitable proxy for the biological potency of light. *Sci Reports.* 2022;12(1). DOI: 10.1038/s41598-022-21755-7.
29. Oh S, Kwak Y, Ohno Y. Vision experiment II on perception of correlated colour temperature. In: *5th CIE Symposium on Colour and Visual Appearance*. Hong Kong. 2020:1-8.
30. Smet KAG, Deconinck G, Hanselaer P. Chromaticity of unique white in illumination mode. *Opt Express.* 2015;23:12488.
31. Rea MS, Freyssinier JP. White lighting. *Color Res Appl.* 2013;38:82-92.
32. Ohno Y, Fein M. Vision Experiment on White Light Chromaticity for Lighting. In: *CIE/USA-CNC/CIE Biennial Joint Meeting*. Kuala Lumpur; 2013.
33. Fotios S. A Revised Kruithof graph based on empirical data. *Leukos.* 2017;13:3-17.
34. Emery KJ, Webster MA. Individual differences and their implications for color perception. *Current Opinion Behav Sci.* 2019;30:28-33.
35. Asano Y, Fairchild MD, Blondé L, Morvan P. Color matching experiment for highlighting interobserver variability. *Color Res Appl.* 2016;41:530-9.
36. Royer M, Murdoch MJ, Smet K, Whitehead L, David A, Houser K, Esposito, T, Livingston J, Ohno Y. Improved method for evaluating and specifying the chromaticity of light sources. *Leukos.* 2023;19:35-52.

Process for Change to an ANSI/IES Standard Under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard Under Continuous Maintenance

NOTE:

Use a separate form for each comment. Submit to the Director of Standards, IES, 85 Broad St, FL 17, New York NY 10004.
Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
Telephone: _____
Fax: _____
E-mail: _____

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____
3. Clause (section), sub-clause or paragraph number; and page number: _____
4. My proposal (check one):
- ☐ Change to read as follows
 - ☐ Delete and substitute as follows
 - ☐ Add new text as follows
 - ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



Lighting Science Standards

Fundamentals, Metrics and Calculations

Lighting Practice Standards

Design, Engineering, and Specifications

Lighting Applications Standards

Design Criteria and Illumination Recommendations

Lighting Measurements and Testing Procedure Standards

Industry Standardization

Roadway and Parking Facility Lighting Standards

Criteria and Illumination Recommendations

Order# ANSI/IES TM-40-24

ISBN 978-0-87995-476-5

www.ies.org